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Demography of woody plants in relation to fire: *Banksia ericifolia* L.f. and *Petrophile pulchella* (Schrad) R.Br.

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Abstract

Plants of Banksia ericifolia *and* Petrophile pulchella *are sensitive to fire. Changes in population size under different fire regimes were estimated, based on measurements of post-fire seedling emergence, seedling survival, survival and seed production in established plants of differing ages, survival of seeds held in serotinous cones and seed-release in the periods between fire.*

Seeds were first available at 5 years in P. pulchella *and 6 years in* B. ericifolia. *Exact replacement would be possible when burnt at these ages, if seedling establishment were very high. Low establishment would delay replacement to 13 years of age in both species. Late summer/autumn fires of high intensity favour high establishment. Such fires at 8–10 year intervals would be tolerated without any sustained decline in numbers. Fires at 10–15 year intervals could occur regardless of season or intensity with little risk of a population decline. Large increases in numbers and density would follow fires spaced at 15–30 years. Enough seeds would be available for replacement up to about 50 years in both species. Viable seed-release in unburnt conditions was sufficient to compensate for deaths in stands over 20 years old, even with very low levels of establishment.*

Two variables accounted for the biggest changes in numbers and density between generations interspersed by fires; namely the age at which a stand is burned and the proportion of seeds which emerge as seedlings. A comparison with other similar species showed similarities in controls on emergence and establishment, lengths of primary juvenile periods and life spans.

Introduction

Plants which rely on seeds for regeneration after fire (obligate seeders) form a prominent part of the vegetation in many fire-prone regions of the world (Gill 1981; Keeley 1981). Some obligate seeders are serotinous as their seeds are retained in the canopy and released by fire (Lotan 1976).

The present report presents demographic studies of two species which co-occur in fire-prone vegetation in the Sydney Region of south-eastern Australia: *Banksia ericifolia* L.f. and *Petrophile pulchella* (Schrad) R.Br. These perennial woody shrubs are frequently dominant in shrublands or the understorey of woodlands. Plants of both species are usually killed by fires of most intensities. Seeds (achenes in *P. pulchella*) are stored in woody infructescences, referred to here as cones, within closed follicles (*B. ericifolia*) or bracts (*P. pulchella*). Many other species of shrubs or small trees in these local vegetation types are also serotinous obligate seeders (Beadle 1940; Clemens & Franklin 1981; Benson 1985).

Crucial aspects of the demography of serotinous, obligate seeders are: (i) the time taken to first reproduction (primary juvenile period, Gill 1975) which determines the minimum possible interval at which fire may recur without causing extinction (Bradstock & Myerscough 1981; Bond *et al.* 1984; Gill & McMahon 1986) because seeds are usually non-dormant; (ii) seedling establishment post-fire, which is determined by rainfall, tempera-

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ture, predators and the rate of release of seeds which, in turn, may be influenced by the season and intensity of fire (Bradstock & Myerscough 1981; Bond *et al.* 1984; Cowling & Lamont 1984, 1985; McMahon 1984); (iii) the size of the pre-fire seed-bank, which is dependent on current storage plus new seeds added after each episode of seasonal fruiting minus losses of stored seeds in existing fruits (McMaster & Zedler 1981; Bond 1984; Gill & McMahon 1986); (iv) rates of survival of seedlings, juveniles and adults; (v) any subsequent establishment in unburnt conditions. Survival will be partly determined by density and age (Schlesinger & Gill 1978; Morris & Myerscough 1983, 1988). If establishment is absent in such conditions, the numbers and density in an even-aged post-fire generation will decline to low levels as the time since last fire nears the limit of longevity of the species (Specht *et al.* 1958; Ashton 1976; Gill & McMahon 1986).

These factors can be linked in a demographic model to predict population size over successive generations subject to re-occurring fires. Such information would provide a detailed basis for the management of fire-prone plant communities. However, few studies have attempted to relate changes in population size to fire regimes (Zedler 1977; Kruger 1983). At present there is much pressure for repeated burning of the vegetation containing the study species at intervals of less than 10 years for the purpose of fuel reduction. Often such prescribed fires are ignited in the cool months, between April and October. If informed fire-management is to take place it is important to know how these fire regimes and their alternatives may affect these potentially vulnerable, serotinous, obligate seeders. This study aimed to answer the following questions.

(1) What is the shortest interval between fires (maximum fire frequency) that either species can withstand for a sustained period (e.g. about 100 years) without suffering a prolonged decrease in population size?

(2) How is this maximum tolerable fire frequency affected by differences in the amounts of seedling emergence and survival occurring after different fires?

(3) Are variations in seedling emergence and survival related to the seasonal timing and intensity of fires?

(4) What are the likely trends in numbers and density of stands when fires occur infrequently?

(5) How do these species compare with other serotinous obligate seeders?

Methods

Study sites

The survival of plants more than 3 years of age, and the sizes of their seed-banks, were studied in seven areas situated within the Brisbane Water National Park, approximately 50 km north of Sydney, from 1980–83. These areas were last burned in 1977, 1976, 1975, 1974, 1973, 1968 and 1960 respectively, determined from fire-history maps and aerial photographs. All were located on ridge-tops of sandy earths derived from Hawkesbury Sandstone. Distance from the coast varied between 3 and 15 km.

Observations on the emergence and survival of seedlings (<3 years of age) were made in populations after four unplanned fires and in one experiment, between December 1980 and November 1982. These were located in the same area and had similar site characteristics to those populations used for the study of established plants.

Seedling emergence and survival

The number of seedlings that emerged after each fire was counted in 40–70 quadrats (0.1 m^2) for each species. The number of pre-fire seeds on dead plants was estimated by counting the number of cones held on branches in transects adjacent to the study quadrats, after the fire. The number of seeds per cone and the proportion of stored seeds that were intact and viable were estimated in cones taken from nearby unburnt stands. These procedures are described below.

The fate of the seedlings was followed until mid 1983. Two alternative models of survival, the exponential and power function (Silvertown 1982) were fitted. No model is given for Burn IV in *P. pulchella* due to the small numbers of plants which emerged. Indeed, models were fitted only when data for major cohorts of 10 or more seedlings were available.

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Emergence was studied in two experiments (commenced May and November 1982) in which artificial ashbeds were prepared and known quantities of seeds sown. Quadrats (1 m×1 m) were prepared by removing all plant cover and litter. Heat was applied to the exposed soil using propane burners. Harvested plant material and litter were ashed and placed on the soil after the heating had ceased. Either 50 (May 1982) or 100 (November 1982) seeds of each species were scattered on the quadrats several days later. The number of emerged seedlings was counted on average every month for a year after the starting dates.

Survival in established plants

Survival was measured in samples of 50 or more plants of both species which were tagged in late 1980–early 1981. Each sample was located within quadrats (1.0 m×1.0 m or 0.5 m×0.5 m) on two adjacent lines randomly situated on a 20 m baseline located in thick patches of either species. All tagged plants were assumed to have arisen as seedlings after the previous fire in each area. The initial height of each plant was recorded.

Plants of *B. ericifolia* in some areas (burned 1977, 1974) were located solely on shallow soils (depth 0.3–0.4 m) and these were stunted in height and spindly in form compared with populations located on deep soils (depth 1.0–1.5 m; burned 1973, 1968, 1960). One area (burned 1975) contained plants on both soils. One sample was tagged in each area on each available soil type. A second was then established in each of the 1968 and 1960 areas, several hundred metres from the first.

Plants of *P. pulchella* were not as common on shallow soils, therefore samples were tagged only on deep soils. These were separate but often close to the *B. ericifolia* samples. No sampling was done in the 1976 or 1973 areas. Two samples were tagged in each of the other areas. A fire in 1982 destroyed a single sample of each species in the 1960 area.

Plants were checked twice yearly from the beginning of 1981 until October 1983 and the proportion of deaths was measured for each area to test for differences in mortality between soils in *B. ericifolia* and between populations of differing age in both species using G tests of independence.

Seed-survival

Ten cones from each of four age classes (1, 5, 9 and 15 years) were harvested from each sample in 1983. These harvests were spread over three to four plants. Ages were estimated either by direct observation (1 and 5 years only) or by position and appearance (9 and 15 years). Cones of 9 and 15 years were those borne nearest the base of plants and presumed to be from the earliest cohort set in the 1968 stand and from the third oldest cohort in the 1960 stand. The latter origin was assumed because some cones in a more advanced state of degradation were found on the ground and these were assumed to be slightly older. In areas where only a single sample was available for each soil type a replicate of ten cones was harvested 50–100 m from the tagged plants. Two samples of cones were also taken from *P. pulchella* plants at the same time as sampling from *B. ericifolia* in the 1973 area.

Counts were made, in the laboratory, of total follicles or bracts and the number that was open and empty was recorded. Cones of *B. ericifolia* were heated in an oven at 200°C for 10 min to open follicles and aid seed removal. This treatment had no effect on seed viability (Bradstock 1985). Cones of *P. pulchella* were left unheated for several weeks after which seed extraction was possible.

Twenty unopened follicles/bracts were selected randomly and examined and the seeds within were scored as damaged by predators or intact. If intact, seeds were classified as filled, desiccated or aborted. Viability of intact, filled seeds was determined in batches of 50 by germination on filter paper in petri dishes maintained at saturation for 2–3 months. Germination occurred under optimum conditions (12:12 h day:night cycle of 15:5°C; Bradstock 1985) in an environmental cabinet.

The relationship between the proportion of seeds per cone surviving (i.e. intact, filled, viable, unreleased seeds) and age was described using various linear and non-linear models. The preferred model was chosen via a comparison of the percentage variances (PV) accounted for. These PV were the differences between the residual and total mean squares expressed as a percentage of the total mean square.

Seed-bank estimation

All new cones set in the samples during the study were counted. Cone production during 1984–85 in the 1960 stands was also measured. Collections of 10 cones produced in two flowering seasons between 1981 and 1983 were made from each sample to determine seed-set per cone. A further 10 cones were harvested in areas with only one sample per soil-depth tested. Mean total annual seed production per plant was thus calculated and then described as a function of age by various linear and non-linear models from which a preferred model was chosen by a comparison of PV.

Seed-bank at any given age (per plant) was predicted by estimating the number of seeds (one seed per follicle or bract) set in each of the successive annual cohorts produced up to that age, using the preferred seed production model. In each cohort survival was then calculated using the preferred survival model for each species. The numbers of surviving seeds from each cohort at a given age were then summed to give an estimate of total stored viable seeds for that age. This was repeated for all ages between the time of first reproduction and 40 years. The estimates above 25 years were therefore extrapolations beyond the age limit of the samples. Allowance was made for the time between flowering and fruit maturity which is about one year in both species.

A further, alternative estimate of seed-bank was made. Seed production was stopped at age 30 years and the attrition of existing seeds was predicted up to an age of 100 years.

Seed-release between fires

The mean number of viable seeds released annually (per plant) in each sample in unburnt conditions was estimated. Samples of cones of four different ages were marked in the 1968 and 1960 samples in mid 1981. Two age classes of cones (1 and 5 years in 1981, method of ageing similar to that described above) were marked in all other samples. The oldest cones in each sample were also marked and in 1981 these were estimated to be 7 and 13 years old. Follicles (*B. ericifolia*) or bracts (*P. pulchella*) which were open and empty were marked. Cones were rechecked after 2 years and the number of follicles/bracts was counted along with those which had opened and shed seeds. Rates of seed-release were compared between samples and cone ages using analysis of variance.

All cones set before 1981 were counted in each sample. The mean number of seeds released per plant was then estimated from the product of the measured annual rate of release per cone age class and the mean number of cones in each class. Cones set before 1981 were allocated to the nearest age class and a correction was made for viability.

The effect of frequent fires

The effect of three successive fires, separated by intervals of identical length, on the number of plants in a hypothetical stand of each species was estimated. Plants living in the first interval were called first generation and in the second, second generation. The first generation immediate pre-fire number was set at 50 and model stands were assumed to be similar to the sample stands in other respects, such as density. Eleven calculations were done, using values between 5 and 15 years for the 'between-fire' interval.

Second generation stand size (N) was calculated as a function of the interval (i) between fires using:

$$N = B \times E \times Ss \times Se \qquad (1)$$

where B = first generation seed-bank size, E = proportion of seeds emerging as post-fire seedlings, Ss = proportion of seedlings which survive and Se = proportion of living plants >3 years old at i years of age. Values of N were divided by 50 to give estimates of proportional change in numbers. These estimates were plotted against size of gap between fires. Opposing, extreme values of E and Ss were used to produce high and low estimates of proportional change.

Results

Seedling emergence and survival

In both species, emergence was spread over periods of up to 9 months after the fires or experiments. The number, density and proportion of seedlings that emerged from seeds were variable (Table 1). Pulses of emergence pro-

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duced cohorts with differing birth dates in each natural fire. In *B. ericifolia* the power and exponential models were found to be the models of best fit (Table 2) while in *P. pulchella* the power function was the model of best fit (Table 2). In some cases neither model fitted the data well (Table 2).

Survival in established plants

Banksia ericifolia: Sample densities are given in Table 3. No significant difference in the overall proportion of deaths was found between samples ($P > 0.10$), on shallow soils. There were too few deaths to allow comparison

TABLE 1. The number, mean density and mean proportion of seeds that emerged as seedlings in *B. ericifolia* (a) and *P. pulchella* (b) after four fires (I–IV) and in two experiments (E1–2)

Date of fire		Number emerged		Density (m^{-2})		Proportion of viable, or planted seeds emerged	
		a	b	a	b	a	b
18/12/80	(I)	182	151	45.5 (2.5)	32.0 (4.1)	0.26	0.26
25/4/81	(II)	75	153	12.5 (1.3)	23.8 (7.3)	0.11	0.06
29/7/81	(III)	96	290	12.5 (5.4)	48.3 (17.1)	0.14	0.10
12/5/82	(IV)	133	20	3.3 (0.5)	3.8 (0.9)	0.27	0.03
15/5/82	(E1)	NA	NA	NA	NA	0.20	0.25
16/11/82	(E2)	NA	NA	NA	NA	0.39	0.33

Standard errors are in parentheses, NA = not applicable.

TABLE 2. Summary of models of best fit, which describe the survival of seedlings which emerged after four fires

Species	Fire	Model	Slope	Intercept	r^2	Half-life (days)
B. ericifolia	I	e	−0.00058	4.47	0.39	967
	II	p	−0.035	4.47	0.75	>1095
	III	e	−0.00055	4.53	0.47	>1095
	IV	p	−0.10	4.28	0.64	33
P. pulchella	I	p	−0.084	4.28	0.33	83
	II	p	−0.073	4.29	0.71	173
	III	p	−0.019	4.54	0.14	>1095

Models fitted were the exponential (e). $\ln N_T = \ln N_0 - bT$, and power function (p), $\ln N_T = \ln N_0 - b \ln T$, where N_0 = the proportional survival at emergence (1.0), N_T = proportional survival at T days after emergence, and b = rate of death.

Fire dates are given in Table 1.

TABLE 3. The initial number and mean density (m^{-2}) of established plants tagged in stands of *B. ericifolia* and *P. pulchella*

Year of last fire	Soil depth	*B. ericifolia*		*P. pulchella*	
		Number	Density	Number	Density
1977	shallow	78	18.5 (5.6)	–	–
1977a	deep	–	–	51	4.5 (0.9)
1977b	deep	–	–	63	3.8 (1.0)
1976	deep	50	6.6 (1.2)	–	–
1975	shallow	55	15.7 (5.2)	–	–
1975a	deep	51	7.1 (1.5)	152	90.6 (22.2)
1975b	deep	–	–	45	9.0 (4.4)
1974	shallow	53	8.5 (4.1)	–	–
1974a	deep	–	–	55	11.0 (3.2)
1974b	deep	–	–	51	14.0 (2.7)
1973	deep	50	3.1 (0.4)	–	–
1968a	deep	50	2.6 (0.6)	50	5.6 (0.8)
1968b	deep	50	2.1 (0.2)	50	5.0 (0.9)
1960	deep	50	1.9 (0.2)	47	2.1 (0.2)

Standard errors are given in parentheses.

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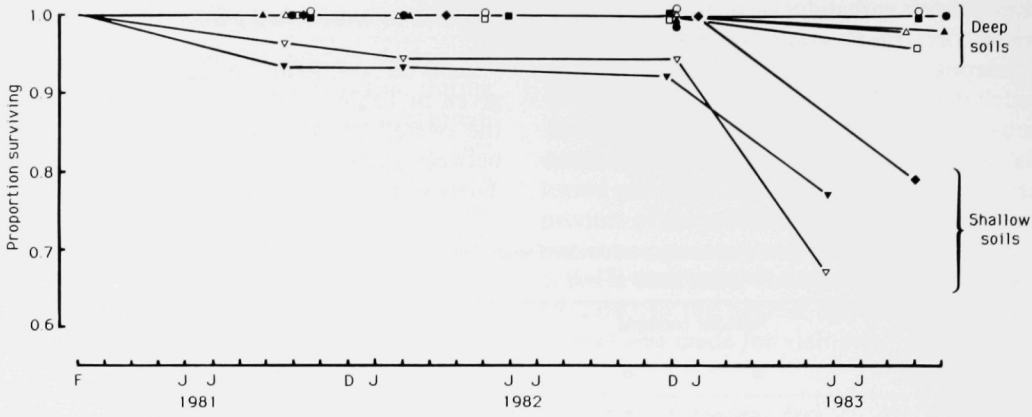


FIG. 1. Survival of established *B. ericifolia* plants. Times of last fire were: deep soils, 1976 (△), 1975 (▲), 1973 (●), 1968a (□), 1968b (■), 1960 (○); shallow soils, 1977 (▼), 1975 (▽), 1974 (◆).

of samples on deep soils, but trends in survival were very similar (Fig. 1). Survival was lower on shallow soils than on deep soils (Fig. 1) but most mortality occurred in 1982–83 on both soil types. Mean death rates (± s.e.m.) on deep and shallow soils were 0.010 ± 0.004 and 0.13 ± 0.016 per annum respectively.

Petrophile pulchella: Sample densities are given in Table 3. Very high mortality was found in the 1974 sample (Fig. 2). In all others there was no significant difference in the proportion of deaths ($P > 0.05$) and the mean death rate, excluding 1974 stands, was 0.031 ± 0.0056 per annum. The pattern of survival (Fig. 2) was similar to *B. ericifolia*, with high mortality in 1982–83, but in *P. pulchella* additional deaths were observed in 1981.

Seed-survival

In both species, the proportion of seeds that were unreleased, intact and viable declined with cone-age (Fig. 3). In *B. ericifolia* this proportion was effectively 1.0 at the time of maturation of follicles (0 years), but in *P. pulchella* at this time about one-third of seeds were viable. Preferred models of proportional survival(s) to year T were:

$s = 0.8663^T$ (2) *B. ericifolia:* PV = 92.6

$s = 0.3021\ (0.9469^T)$ (3) *P. pulchella:* PV = 71.6

Half-lives of seeds were 4.8 and 13.7 years respectively.

Seed-bank estimation

The age at first flowering was 4 years in *P. pulchella* and 5 years in *B. ericifolia* on deep soils during this study. No cones were set in *B. ericifolia* on shallow soils though a few inflorescences were produced in adjacent untagged plants.

Substantial differences occurred between years and samples, in both the numbers of cones produced and the seeds set within them. These produced large variations in seed production (Fig. 4). There was no evidence of any serial correlation between data from successive years for individual samples.

In *P. pulchella*, seed-production was very high in the 1960 sample (Fig. 4). Most cones were young which indicated that cone-production may accelerate rapidly between the ages of 15 and 20 years.

The preferred models of seed-production per mean plant (n) at stand age (T) were:

$n = 4.408\ (1.116^T)$ (4) *B. ericifolia:* PV = 70.5

$n = 7.418\ (1.117^T)$ (5) *P. pulchella:* PV = 65.1

Predictions of seed-bank size are given in Fig. 5.

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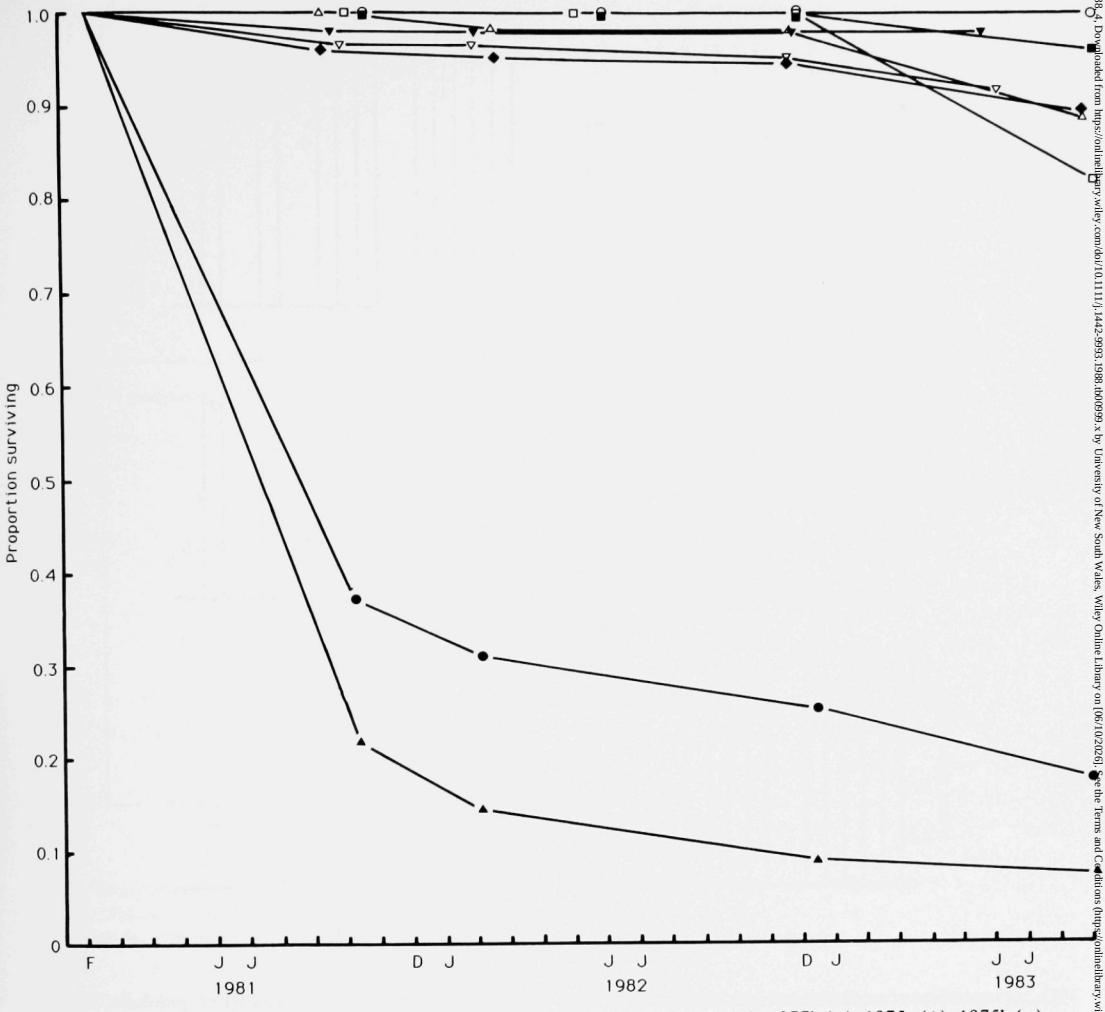


FIG. 2. Survival of established *P. pulchella* plants. Times of last fire were: 1977a (▼), 1977b (▽), 1975a (♦), 1975b (△), 1974a (▲), 1974b (●), 1968a (□), 1968b (■), 1960 (○).

Seed-release in the interval between fires

A comparison of seed-release rates indicated that two pooled estimates could be made in *B. ericifolia* (1–5 years, 0.021, and 7–13 years, 0.011 seeds/cone per annum) and three in *P. pulchella* (0.009, 0.036 and 0.041, for 1, 5 and 7–13 year old cones respectively). Estimates of mean number of seeds released annually in the four oldest samples are given in Fig. 6.

The effect of frequent fires

The alternative values of E were set at 0.5 and 0.05. These bounded the range observed in the field (Table 1). In glasshouse studies, levels of emergence in excess of 0.5 were observed in both species (Bradstock 1985), but elsewhere in the field, in *B. ericifolia* only (Bradstock & Myerscough 1981). The survival models from the April 1981 and December 1980 (highest survival), and May 1982 and July 1981 (lowest survival) fires were used to calculate values of Ss in *B. ericifolia* and *P. pulchella* (Table 2) respectively. Values of Se were estimated from the appropriate mean death rates.

The estimates of between-fire intervals at which replacement was possible in second generation stands were (Fig. 7) 5 and 6 years (high E and Ss, *P. pulchella* and *B. ericifolia* respec-

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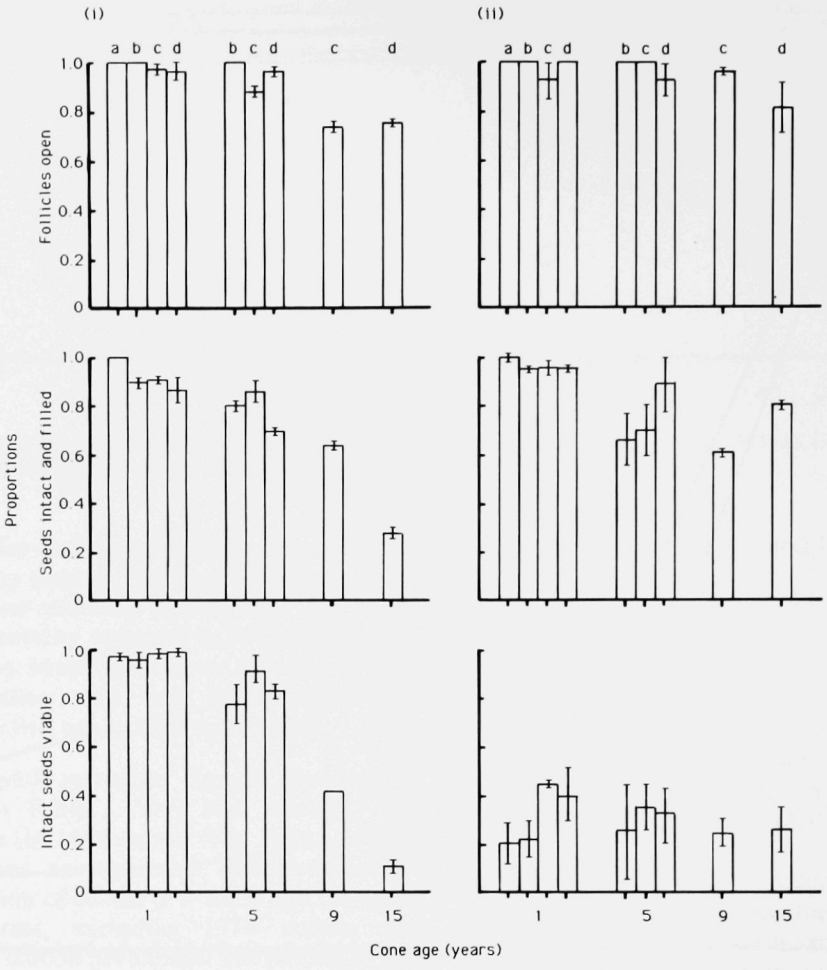


FIG. 3. Proportions of seeds retained intact and viable in cones of differing ages in (i) *B. ericifolia* and (ii) *P. pulchella*. Times of last fire were: a = 1976, b = 1973, c = 1968, d = 1960.

tively) and 13 years (low E and Ss, both species). Fires at longer intervals than these would cause growth in the size of populations under the specified conditions of emergence and survival (Fig. 7).

Discussion

Effects of frequent fires

The importance of emergence and survival was reflected in the variability of the interval between fires at which stand replacement would be possible (Fig. 7). Burning at an interval close to 13 years could be done with little regard to other factors such as season or intensity of fire without risk of a decline in numbers. The low levels of emergence and survival (Tables 1, 2) used to produce this estimate were the product of a severe drought in the area (Bureau of Meteorology, pers. comm.). It is likely that fires in most years will produce at least a similar or larger number of seedlings. Survival of established plants may also be higher, for this reason.

Fire-managers may wish to burn as often as twice a decade, hence it is necessary to know the limit to which fire frequency can be reduced below 13 years without risking a serious decline in numbers. Seasons and intensities of fire may become important in such

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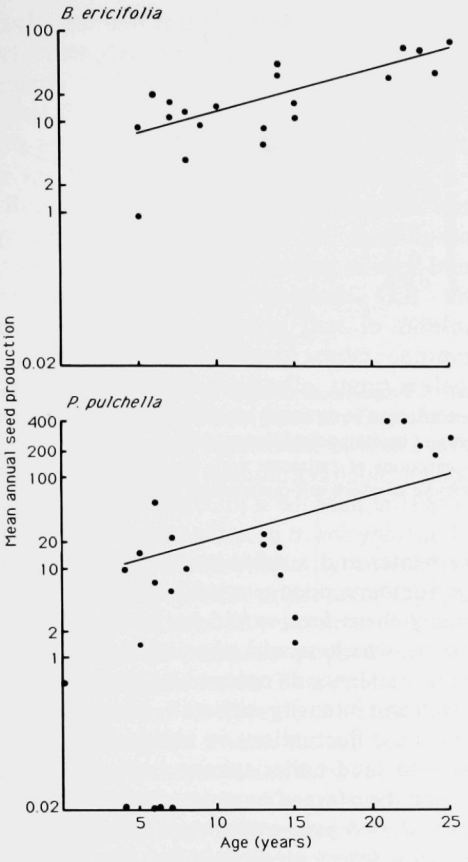


FIG. 4. Mean annual seed-production per plant as a function of age. Preferred models (highest percentage variance accounted for) are indicated by unbroken lines.

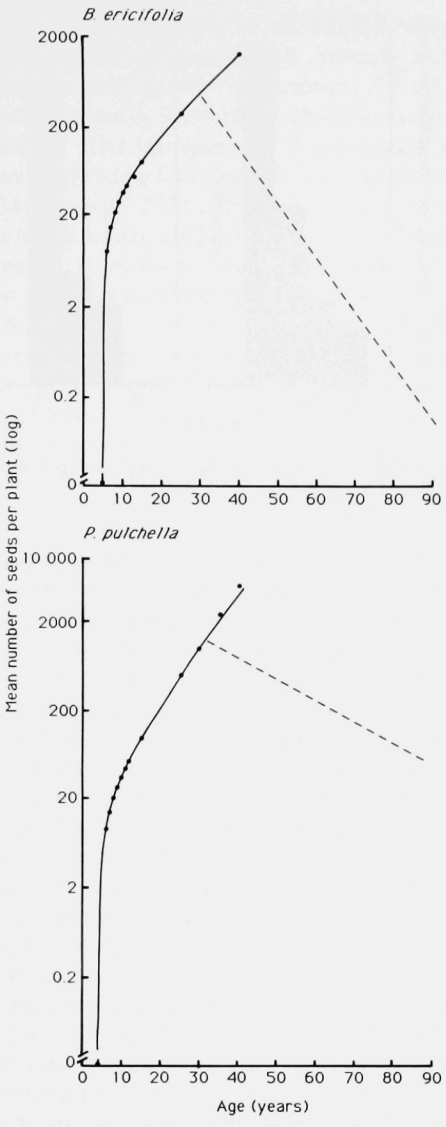


FIG. 5. Mean number of stored viable seeds (seed-bank) per plant, under two regimes of seed production: continuous (——), ceased at 30 years (– – –).

circumstances as will year to year variability in rainfall.

Bradstock (1985) found that emergence in *B. ericifolia* and *P. pulchella* varied with rainfall, soil moisture and temperature. High summer temperatures impose a form of exogenous dormancy on seeds (definition of Mott & Groves 1981), restricting emergence to the cooler months. Seeds which lie on the soil over summer lose some viability. The amount of emergence was positively correlated with the period of time the soil surface (0–1.0 cm) remains at field capacity after rain. Seed predation was minimal in burnt areas.

Rainfall near the coast in the Sydney Region is highest, on average, from January to July (100–130 mm/month), and lowest from August to December (70–80 mm/month); however, there is great year to year variation. When evaporation is taken into account, the average period of sustained high soil moisture is May–July (early winter). Emergence may therefore be highest after fires which are timed so that most seeds are released in early winter. Seed-release will be most rapid after high intensity fires (Bradstock & Myerscough 1981; Bradstock 1985) and such fires in late summer–autumn will cause most seeds to be released by early winter. In contrast, low inten-

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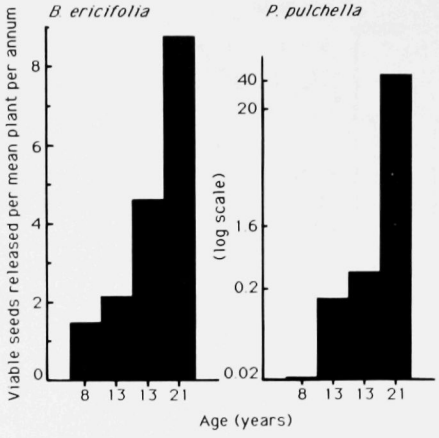


FIG. 6. Estimated annual release of viable seeds per plant in samples of differing age.

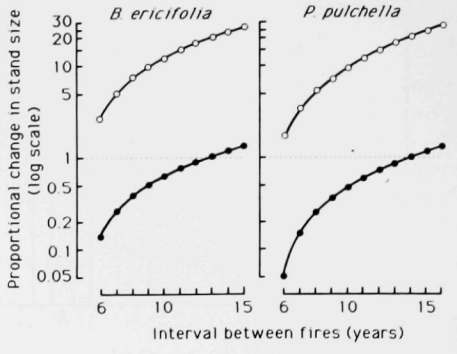


FIG. 7. Proportional changes in sizes of stands over two generations of equal length as a function of interval between fires and seedling establishment (maximum ○; minimum ●). Intersections of estimates with the broken line indicate intervals at which self-replacement is possible (see text).

sity fires at this time will prolong seed-release over winter and spring. Late summer–autumn, high intensity fires will therefore, on average, yield optimum emergence. A similar fire regime giving optimum emergence in other species of serotinous Proteaceae has been observed by Specht *et al.* (1958), Specht (1981), Bond *et al.* (1984), Cowling and Lamont (1984a, 1984b, 1985), McMahon (1984). Mechanisms in each case were generally similar to those described here.

While mortality of plants older than 100 days was due mainly to drought, younger seedlings died during both short dry spells and heavy rain (Bradstock 1985) when they were broken by raindrop impact or washed away. Bradstock (1985) concluded that the risk of death at the most vulnerable age (<100 days; Table 2) would be minimized when seedlings emerged in early winter. This reinforced the conclusion that late summer fires are optimal in producing seedlings.

After successive fires at 8 years, the proportional change in second generation stand size may vary from 9.8 to 0.5 in *B. ericifolia* and from 7.2 to 0.6 in *P. pulchella* (Fig. 7). The daily rainfall patterns for Sydney over 50 years (Bureau of Meterology, pers. comm.) show that a winter containing two successive months of low soil surface moisture occurs on average once every decade whereas very wet winters in which the soil is more or less continually moist occur several times a decade (Bradstock 1985). Thus fires of optimal season and intensity occurring 8 years apart would be followed by a dry winter and a subsequent decline in post-fire numbers, once every 80–100 years. More usually these fires would be followed by wet winters which would permit an increase in numbers. Hence an optimal regime in terms of season and intensity with an 8–10 year interval may cause fluctuations in numbers but is unlikely to send either species into a persistent decline if sustained over a period of up to 100 years. A 6–8 year cycle would be hazardous, due to a lower seed-bank and the increased probability of encountering dry post-fire conditions.

Reproduction is delayed in *B. ericifolia* growing on shallow soils and such stands may decline when fires occur less than 15 years apart. Observations indicate that the primary juvenile period may extend to more than 8 years and also that the seed-bank size may be less than on deep soils.

There are, therefore, two consecutive phases in which plants are susceptible to recurrent fires: first, during the primary juvenile period and second, the subsequent time up to the age at which exact self-replacement of stands is possible. So far most attention has been given to describing the lengths of primary juvenile periods. Values given here fell within the ranges already reported for these species (Benson 1985) and for other serotinous obligate seeders of the Proteaceae (3–11 years; Kruger 1983; Lamont 1985). Measurements of this second critical phase need to be made if serotinous obligate seeders are to be used as indicator species in fire-prone vegetation. It

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may be useful if correlations exist between this and other demographic characteristics, particularly the length of the primary juvenile period. For example, is first reproduction at an early age always followed by a short interval to the age of self-replacement or vice versa?

Replacement ages for the study species were similar, despite differences in seedling-survival, early rates of development of seed-bank and primary juvenile period. Gill and McMahon (1986) reported that in *Banksia ornata*, a serotinous obligate seeder, common in parts of southern Australia, stand replacement may be likely after 16 years of age if post-fire emergence took place from 20% of available seeds. Seeds first become available at about 6 years in *B. ornata* (Gill & McMahon 1986), a figure comparable to that in *B. ericifolia* and *P. pulchella*. Despite this similarity, populations of *B. ornata* may decline under some repeated cycles of fire less than 15 years apart, whilst *B. ericifolia* and *P. pulchella* populations may increase or remain stable. This difference was primarily due to a slower rate of accumulation of seed-bank in *B. ornata*, though other possible differences in projected survival rate were also present. Similarly, Kruger and Bigalke (1984) reported that in *Protea repens*, a shrub of the South African fynbos, the primary juvenile period may be 4–5 years (ripe seeds first available), but replacement age may be 10–12 years. It is most prudent to conclude that there is little current evidence of any consistent linkage between these key events. Further demographic studies amongst a wider range of species are needed to confirm this.

Effects of infrequent fires

The present study was hampered by a lack of old populations (>25 years). This prevented direct measurements of life span and seed-production in very old plants. It was necessary to extrapolate trends from younger populations and anecdotal evidence to describe what happens when fire is excluded for long periods.

In the seed-bank models (Fig. 5) seed-production was arbitrarily set to cease at 30 years. However, Bradstock and Myerscough (1981) observed the occurrence of stands of *B. ericifolia* older than this (these also contained *P. pulchella*) in which seeds were still being produced. The modeled storages will therefore underestimate the trend in real stands. Nevertheless, these suggested that enough seeds would be available for replacement for up to about 45 years in *B. ericifolia* and 90 years in *P. pulchella*. The life spans of these species are more likely to be less than 50 years (Bradstock & Myerscough 1981), hence in reality, senescence will hasten the decline of the seed-bank through the death of plants and replacement ages may be much nearer 30 years. The fall of viable seeds onto the litter and soil will increase as a consequence and, if seed-production occurs beyond 30 years, this pool of released seeds will be very large. If these assumptions are correct, changes in number and seed-bank may be quite similar in their pattern to those described by Gill and McMahon 1986 for *B. ornata*. Kruger and Bigalke (1984) also reported a similar decline in numbers after 30 years in some serotinous obligate seeders from South African shrublands, and suggested that life spans for dominant species of this kind fell within the range 20–50 years.

In samples that are more than 20 years of age the number of viable seeds released each year (Fig. 6) greatly exceeded the number of plants that died annually (Figs 1, 2). Bradstock (1985) found that the heavy cover of plants and seedling predators restricted establishment from this pool of available seeds. However, a few seedlings are likely to survive in both species. A rough calculation can be done to estimate the number which may establish in a stand of 50 plants. Average conditions of soil moisture in any year will usually be adequate to give 10% emergence of available seeds. If 5% of these survive for 3 years (about eight times lower survival than in burnt conditions; Table 2), then 21–24 year old stands would produce two (*B. ericifolia*) and 12 (*P. pulchella*) seedlings annually. Such recruits would join the ranks of small suppressed plants which are subject to high mortality and would probably have little chance of reproducing (Bradstock 1985).

At greater ages, when seed-release and mortality may be high, gaps in the canopy caused by senescence of dominants may appear. Seedling predators may become satiated as a result (e.g. Bond 1984a; O'Dowd & Gill 1984) and these conditions may promote the emergence, growth and reproduction of new plants. The relative abilities of *B. ericifolia*, *P. pulchella*

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and other species to exploit gaps and openings may determine whether self-perpetuation in large numbers is possible in the absence of fire. More work is needed to determine regeneration processes in older stands and their scarcity means that some effort needs to be made to preserve unburnt areas in the future.

Fires, 20–30 years apart, will create very high density stands and the absence of these in the study area may be due to a history of burning at a higher frequency.

Effects of density

Fire regimes may be used to increase or decrease the density of stands. For instance, if the relationships in Figs 1–7 hold true for much of the potential range of densities, then it may be possible, for example, to thin very dense stands and then hold them in this state. Several adverse fire-cycles (e.g. low intensity spring fires, less than 10 years apart) followed by a 10 year cycle of high intensity autumn fires would achieve this result.

The density of a stand will change with age as plants die and the chronosequence in Table 3 may indicate the approximate trend. Bradstock (1985) found that density had no effect on the survival of seedlings older than 100 days. There were also no obvious effects of density on survival at older ages in either species, including the 1975 samples in *P. pulchella* (Fig. 2; Table 3), although dead plants tended to be small suppressed individuals in both species. Morris and Myerscough (1983, 1988) found evidence for density-dependent mortality in *B. ericifolia* stands of very high density (initial sample: 140 plants m^{-2}) between the ages of 4 and 9 years.

In both *B. ericifolia* and *P. pulchella*, the proportion of plants that continues to produce cones declined with age (Bradstock 1985) so that over time the increase in reproductive effort (Fig. 4) came from a small number of dominant individuals (Bradstock 1985). This pattern of reproduction amongst individuals may affect the relationship between density and seed-bank if the survival of dominant reproductive plants is unaffected by the number of suppressed individuals. The number of plants which reproduce and the amount of seeds produced may then be fixed above a moderate threshold of density, as indicated by experimental studies in other single-species populations (Harper 1977; Silvertown 1982). Further evidence of Morris and Myerscough (1988) showed that *B. ericifolia* plants at densities of 27.5 plants m^{-2} had set a mean of 345 ± 92 seeds m^{-2} at 8 years whereas high density plants (142 m^{-2}) produced 225 ± 80 seeds m^{-2}. If seed-bank size is fixed across the upper range of densities then its prime determinant will be stand age. The perpetuation of very dense stands would then be possible only after longer fire-free intervals than those described here for lower density samples.

Density and numbers may also be influenced by pollinators. Paton and Turner (1985) found that introduced bees may double the numbers of seeds set in *B. ericifolia* cones. The introduction of exotic bees may therefore have caused a massive increase in density and numbers in stands containing this species without any concurrent change in fire frequency. Other changes in species composition may have followed.

Conclusions

All components of fire regime can influence a change in numbers and density, between generations, of *B. ericifolia* and *P. pulchella*. However, the largest change was due to differences in the age at which stands were burnt and the proportion of seeds which emerged as seedlings. Seasonal controls on seedling establishment, lengths of primary juvenile periods, effects of seed or seedling predators in unburnt conditions, and changes in structure of old stands and life spans, are generally similar in southern hemisphere species, particularly within the family, Proteaceae. However, demographic behaviour of fire-prone species cannot be predicted solely from any single characteristic such as survival, degree of serotiny (seed-survival in cones), seed-bank size or success of emergence. Any attempt to compare and classify species on their relative ability to withstand specific fire regimes, without a quantitative overview of the life-cycle, could be misleading.

There may be little reason to assume that if reproduction commences early, then enough seeds will also be available to ensure replacement at a relatively early age. Quantitative variations in all the important components of the

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life-cycle may exist within a species and correlations between these components and obvious site factors need to be studied (e.g. Cowling & Lamont 1985). Managers concerned with conserving populations at stable numbers should therefore use, as an indicator of the minimum interval between fires, the serotinous obligate seeder within their estate, that is slowest to reach self-replacement.

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